

Ferrite core and transformer — Solution

X24 (2024) — English translation

A0

?? pts

Measure the resistance of coils A and B (r_A and r_B respectively), as well as the resistance of the resistors connected in series with them. All values must be in the range up to 15 Ω .

Using a multimeter in ohmmeter mode, we measure the resistance of the coils and make sure they are working.

Answer: \textbf{Answer:}

$$r_A = (13.3 \pm 0.1) \Omega$$

$$r_B = (9.0 \pm 0.1) \Omega$$

$$r_{A,} = r_{B,} = (0.97 \pm 0.04) \Omega$$

A1

0.60 pts

Carry out measurements to verify the above model of magnetic field shielding at frequency $f = 10$ kHz. Plot a linearized graph and find the damping coefficient α for the given frequency. The foil must be placed between coil A and the sensor coil.

From Faraday's law of electromagnetic induction it follows that the induced emf

$$\mathcal{E}(t) = -N \frac{d\Phi(t)}{dt} = -NS \frac{dB}{dt}$$

Neglecting омическим resistanceм катушки, we get, что voltage на катушке

$$U(t) = -\mathcal{E}(t) = NS \frac{dB}{dt}$$

Consequently, амплитуды напряжения на катушке and индукции магнитного поля связаны как

$$U = 2\pi f_0 NSB \Rightarrow U \sim B$$

Let U_0 - amplitude напряжения на катушке датчика в отсутствии фольги, U_d - amplitude напряжения на катушке датчика for/at слое фольги толщиной d m. B_0 and B_d we determine similarly. Then

$$\frac{B_d}{B_0} = \frac{U_d}{U_0} \Rightarrow -\alpha d = \ln \frac{B_d}{B_0} = \ln \frac{U_d}{U_0}$$

It can be seen that $\ln \frac{U_d}{U_0}(d)$ - linear dependence The table of measured and calculated values is presented below.

Neglecting the ohmic resistance of the coil, we obtain that the voltage across the coil is

$$U(t) = -\mathcal{E}(t) = NS \frac{dB}{dt}$$

Consequently, the amplitudes of the voltage on the coil and the magnetic field induction are related as

$$U = 2\pi f_0 NSB \Rightarrow U \sim B$$

Let U_0 be the voltage amplitude on the sensor coil in the absence of foil, U_d be the voltage amplitude on the sensor coil with a foil layer of thickness d μm . B_0 and B_d are defined similarly. Then

$$\frac{B_d}{B_0} = \frac{U_d}{U_0} \Rightarrow -\alpha d = \ln \frac{B_d}{B_0} = \ln \frac{U_d}{U_0}$$

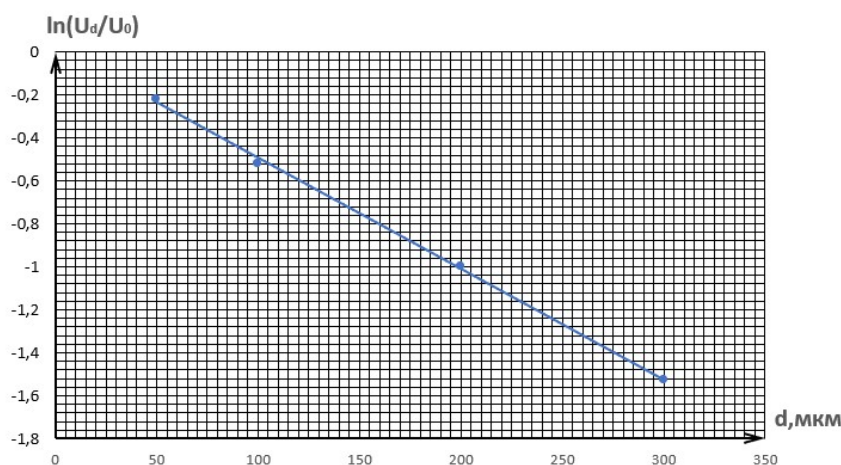
It can be seen that $\ln \frac{U_d}{U_0}(d)$ is a linear dependence

A table of measured and calculated values is presented below.

$d, \mu\text{m}$ U_d, B $\ln \frac{U_d}{U_0}$ 0 2.12 0 50 1.70 -0.22 100 1.26 -0.52 200 0.78 -1.00 300 0.46 -1.53

Let's plot the linear dependence $\ln \frac{U_d}{U_0}(d)$ and determine α from the slope angle coefficient.

Answer: \textbf{Answer: } $\alpha = 5160 \text{ m}^{-1}$



A2

2.00 pts

Determine the attenuation coefficient α for several frequencies in the range 5 – 25 kHz.

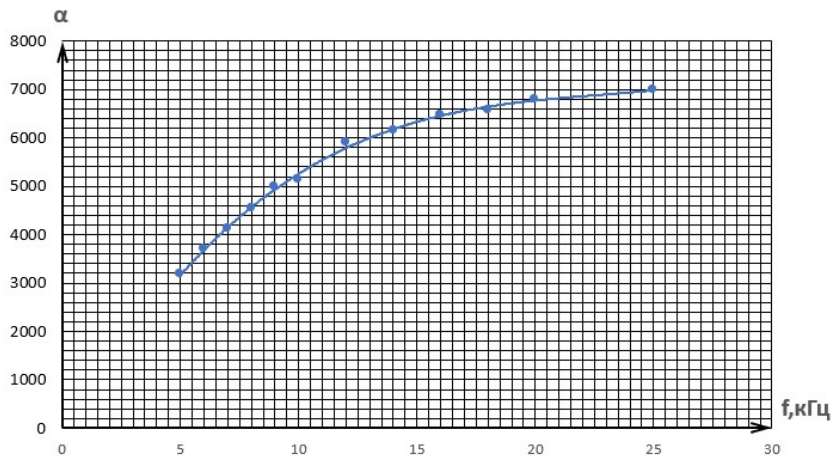
Let's repeat the measurements from the previous point for other frequencies. The measurements and their corresponding α are presented in the table below.

f, kHz	U_0, V	U_{50}, V	U_{100}, V	U_{200}, V	U_{300}, V	$\alpha, 10^3 \text{m}^{-1}$
5.00	2.10	1.88	1.62	1.17	0.74	3170
6.00	2.10	1.84	1.55	1.06	0.65	3690
7.00	2.12	1.82	1.48	0.98	0.58	4130
8.00	2.12	1.78	1.41	0.90	0.53	4540
9.00	2.12	1.74	1.34	0.82	0.50	5000
10.00	2.12	1.70	1.26	0.78	0.46	5140
12.00	2.12	1.62	1.14	0.66	0.40	5910
14.00	2.21	1.54	1.03	0.60	0.34	6160
16.00	2.15	1.46	0.94	0.54	0.31	6480
18.00	2.31	1.38	0.86	0.50	0.30	6580
20.00	2.18	1.31	0.82	0.46	0.27	6810
25.00	2.58	1.04	0.70	0.40	0.25	6980

A3

1.20 pts

Plot a graph of α versus frequency.



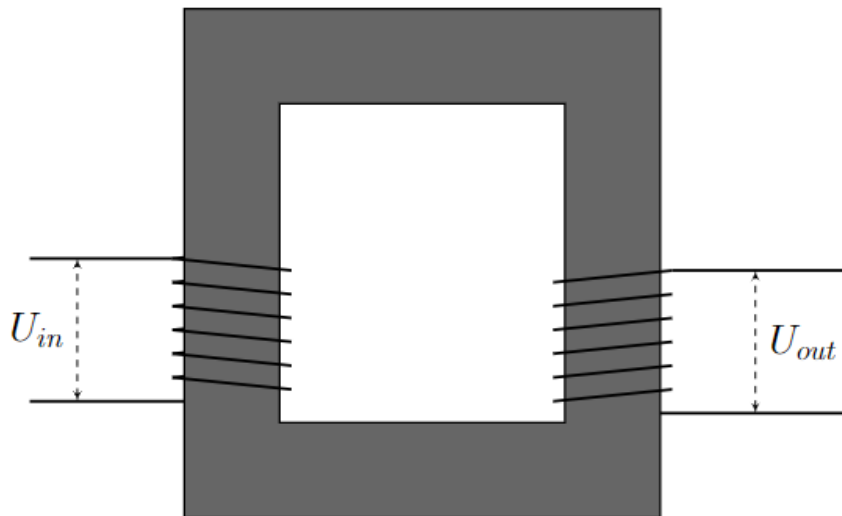
B1

0.30 pts

Determine m experimentally. Draw the measurement diagrams necessary to obtain the numerical value of this quantity.

Let's assemble the circuit shown in the figure and measure U_{in} and U_{out} using an oscilloscope. $U_{in,0} = (6,10 \pm 0.05)$ V. $U_{out,0} = (3.48 \pm 0.04)$ V. We get the value $m = \frac{U_{out,0}}{U_{in,0}}$:

Answer: $m = 0.57 \pm 0.01$.



B2

1.30 pts

Remove the dependence of P and P_0 on the current amplitude in the secondary coil $I_{B,0}$.

Let's introduce the following notations: U_1 and I_1 are the amplitudes of voltage and current in the primary coil, respectively, φ is the phase difference between the voltage and current in the primary coil, U_2 is the amplitude of the voltage on the rheostat, I_2 is the amplitude of the current through it ($I_2 = I_{B,0}$). Then $P_0 = \frac{U_1 I_1 \cos \varphi}{2}$, $P = \frac{U_2 I_2}{2} = \frac{U_2^2}{2R}$. U_1 , U_2 , φ are measured directly using an oscilloscope. $I_1 = \frac{U_r}{r}$, where $r = 10 \Omega$ is the resistance connected in series with the primary coil, U_r is the voltage amplitude across it. $I_{B,0}$ changes when the rheostat knob is turned. The measurement and calculated values

are shown in the table below.

U_1, V	I_1, mA	$\cos \varphi$	P_0, mW	R, Ohm	U_2, V	$I_{B,0}, mA$	P, mW	η
3.36	59.0	0.85	84.1	2.1	0.16			
78.1	6.4	0.08	3.60	54.5	0.88	86.6	8.3	0.58
70.0	20.3	0.23	4.20	39.2	0.92	75.8	33.5	1.77
52.9	46.9	0.62	4.60	31.8	0.91	66.9	53.8	2.26
42.0	47.4	0.71	4.92	26.2	0.88	56.9	79.3	2.62
33.0	43.1	0.76	5.10	23.0	0.86	50.3	102	2.83
27.8	39.3	0.78	5.20	21.8	0.77	43.3	125	2.89
23.2	33.4	0.77	5.30	19.4	0.80	41.1	146	3.09
21.2	32.7	0.80	5.45	16.6	0.71	32.0	200	3.26
16.3	26.5	0.83	5.60	15.6	0.55	23.8	303	3.45
11.4	19.6	0.82	5.70	14.0	0.50	20.0	399	3.55
8.9	15.7	0.79	5.70	13.2	0.44	16.5	506	3.56
7.0	12.5	0.76	5.85	12.8	0.28	10.3	1080	3.65
3.4	6.2	0.60	5.90	13.0	0.14	5.3	5810	3.81
0.7	1.2	0.23	5.90	12.8	0.14	5.2	10500	3.85
0.13								

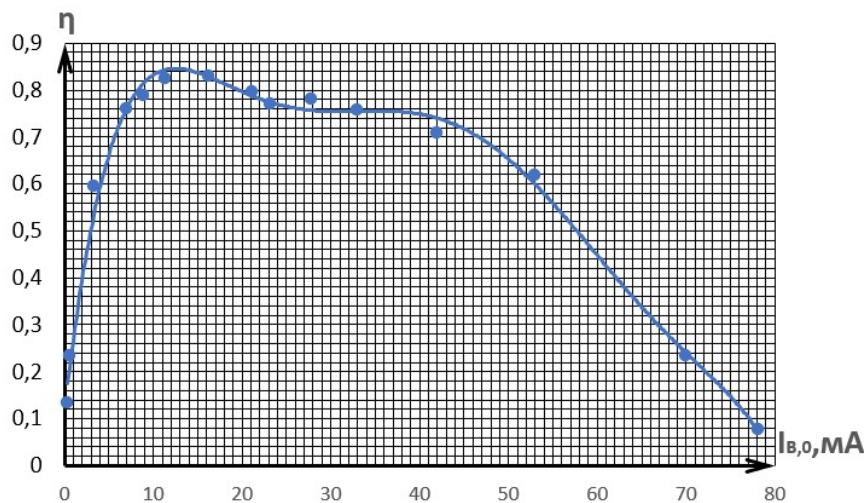
B3

1.20 pts

Plot the dependence $\eta(I_{B,0})$. Indicate the maximum efficiency η_{max} and at what $I_{B,0}^{max}$ it is achieved.

From the graph we find η_{max} and $I_{B,0}^{max}$:

Answer: \textbf{Answer: } $\eta_{max} = 0.83 \pm 0.01$. $I_{B,0}^{max} = (15 \pm 5) \text{ mA}$.



B4

2.00 pts

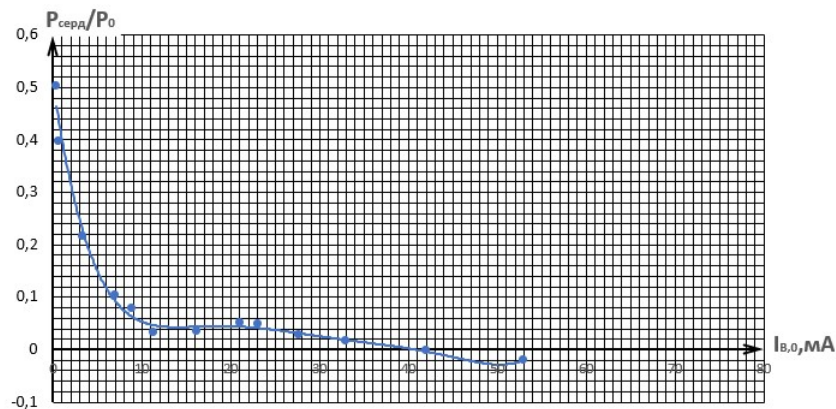
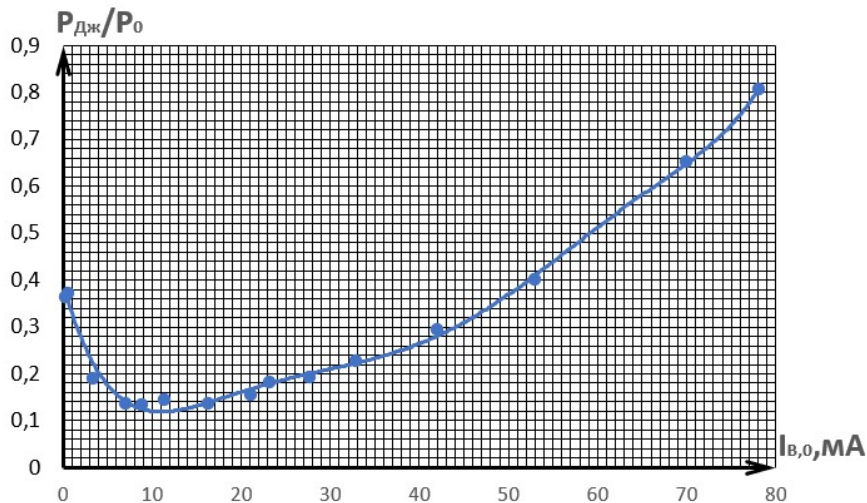
Obtain the dependences of the relative contribution of each loss source $\frac{P_J}{P_0}$ and $\frac{P}{P_0}$ on the current amplitude in the secondary coil $I_{B,0}$. Graph them.

Joule losses consist of the power dissipated directly on the coils due to the presence of ohmic resistance, and the power dissipated by a resistor connected in series with the primary coil. Knowing the currents in both coils, their resistances and the value of the resistor r , we can calculate the Joule losses P_J , and the losses in the core P will be found as the remainder of the "expansion" equation P_0 in the condition.

$$P_J = I_1^2(r_A + r) + I_2^2 r_B,$$

$$P = P_0 - P_J.$$

$I_{B,0}, mA$	P_J, mW	P, mW	$\frac{P_J}{P_0}$	$\frac{P}{P_0}$
78.1	67.7	10.0	0.81	0.12
70.0	56.4	9.9	0.65	0.11
52.9	30.4	0	0.40	0
42.0	19.6	0	0.29	0
33.0	12.8	1.0	0.23	0.02
27.8	9.6	1.4	0.19	0.03
23.2	7.9	2.1	0.18	0.05
21.2	6.4	2.0	0.16	0.05
16.3	4.4	1.1	0.14	0.03
11.4	3.4	0.8	0.14	0.03
8.9	2.6	1.6	0.13	0.08
7.0	2.3	1.7	0.14	0.10
3.4	2.0	2.2	0.19	0.21
0.7	2.0	2.1	0.37	0.40
0.13	0.4	1.9	0.26	0.36
0.50				



C1

0.10 pts

Obtain an equation that relates the amplitudes of current and voltage in the coil.

From Faraday's law of electromagnetic induction, induced emf

$$\mathcal{E}(t) = -N \frac{d\Phi(t)}{dt} = -\mu\mu_0 g N^2 \frac{dI(t)}{dt}$$

В связи с пренебрежением омическим сопротивлением катушке

$$U(t) = -\mathcal{E}(t) = \mu\mu_0 g N^2 \frac{dI(t)}{dt}$$

It immediately follows

Answer: \textbf{Answer:}

$$U = 2\pi\mu\mu_0 g f_0 N^2 I$$

C2

0.10 pts

In this case, obtain expressions for the induced emf $\mathcal{E}_A(t)$ and $\mathcal{E}_B(t)$ in coils A and B, respectively. Express your answer using N_A , N_B , k , $U_g(t)$. Consider the current in the primary coil to be positive if it flows from the "plus" of the source to the "minus".

Due to the neglect of the ohmic resistance of the coils

$$-N_A \frac{d\Phi_A}{dt} = \mathcal{E}_A(t) = -U_g(t),$$

$$\mathcal{E}_B(t) = -N_B \frac{d\Phi_B}{dt} = -kN_B \frac{d\Phi_A}{dt} = -\frac{kN_B}{N_A} U_g(t).$$

Answer: $\mathcal{E}_A(t) = -U_g(t)$, $\mathcal{E}_B(t) = -\frac{kN_B}{N_A} U_g(t)$.

C3

0.30 pts

Obtain an expression for the derivative of the current in coil A $\frac{dI_A(t)}{dt}$. Express your answer using μ , μ_0 , g , N_A , N_B , k , $U_g(t)$, $I_B(t)$. Consider currents I_A and I_B to be of the same sign if the currents they create add up.

$$\mathcal{E}_A(t) = -N_A \frac{d(\Phi_A + k\Phi'_B)}{dt} = -\mu\mu_0 g N_A \left(N_A \frac{dI_A(t)}{dt} + kN_B \frac{dI_B(t)}{dt} \right) = -U_g(t).$$

From here we get

Answer:

$$\frac{dI_A(t)}{dt} = \frac{1}{\mu\mu_0 g N_A^2} \left(U_g(t) - \mu\mu_0 g k N_A N_B \frac{dI_B(t)}{dt} \right).$$

C4

0.60 pts

Obtain theoretical expressions for the self-inductances L_A and L_B . Express your answer using μ , μ_0 , g , N_A , N_B .

Using Faraday's law of electromagnetic induction and the expression for the magnetic field flux, we obtain:

$$\mathcal{E}(t) = -N \frac{d\Phi(t)}{dt} = -\mu\mu_0 g N^2 \frac{dI(t)}{dt}$$

И from the definition индуктивности we find выражения for L_A and L_B :

Answer:

$$L_A = \mu\mu_0 g N_A^2,$$

$$L_B = \mu\mu_0 g N_B^2.$$

C5

0.80 pts

Determine L_A , L_B , k experimentally. Draw the measurement diagrams needed to obtain numerical values for these quantities.

Let us determine the amplitudes of voltage and current in coil A U_A , and I_A . The voltage amplitude is directly measured by an oscilloscope; to determine the current amplitude, you need to measure the voltage amplitude across the resistance connected in series to the coil (see figure) and divide it by the resistance value.

$$U_A = (6.0 \pm 0.1) \text{ V}, \quad I_A = (1.20 \pm 0.05) \text{ mA}.$$

Then we use expression for импеданса катушки

$$L_A = \frac{U_A}{2\pi f_0 I_A} = (80 \pm 5) \text{ mH}$$

Similarly for катушки B

$$U_B = (6.0 \pm 0.1) \text{ V}, \quad I_B = (2.80 \pm 0.15) \text{ mA}.$$

$$L_B = \frac{U_B}{2\pi f_0 I_B} = (34 \pm 2) \text{ mH}.$$

Чтобы олмитандъ k нужно найти ratio induced EMF катушек в расчёте на один wand-current:

$$k = \frac{U_{out,0}/N_B}{U_{in,0}/N_A} = m \frac{N_A}{N_B} = 0.86 \pm 0.02.$$

Then we use the expression for the coil impedance

$$L_A = \frac{U_A}{2\pi f_0 I_A} = (80 \pm 5) \text{ mH}$$

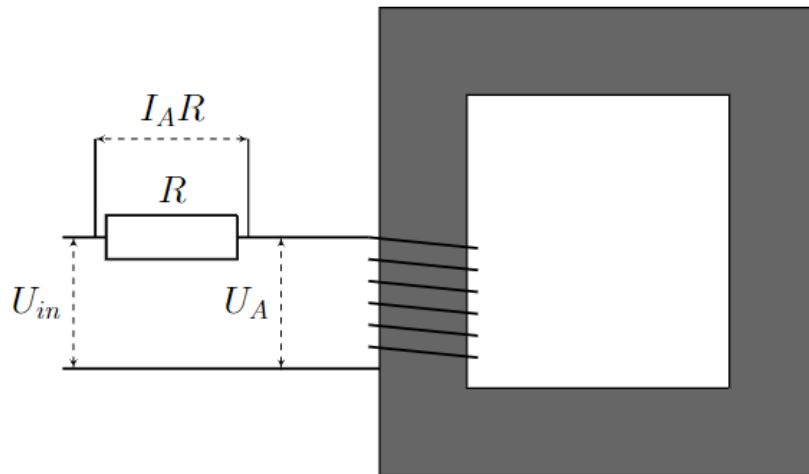
Similarly for coil B

$$U_B = (6.0 \pm 0.1) \text{ V}, \quad I_B = (2.80 \pm 0.15) \text{ mA}.$$

$$L_B = \frac{U_B}{2\pi f_0 I_B} = (34 \pm 2) \text{ mH}.$$

To determine k you need to find the ratio of the induction emf of the coils per turn:

$$k = \frac{U_{out,0}/N_B}{U_{in,0}/N_A} = m \frac{N_A}{N_B} = 0.86 \pm 0.02.$$



C6

0.30 pts

Show that neglecting the internal resistance of the coil is justified.

The internal resistance of the coil is $r_{A,B} \approx 10 \Omega$. The impedance of its inductance at frequency f_0 $Z_{A,B} \approx 3000 \Omega$. It follows that $r_{A,B} \ll Z_{A,B}$, therefore neglect is justified.

C7

0.60 pts

Obtain an explicit expression for $I_P(t)$ in terms of f_0 , L_A , k , $U_g(t)$ and find the numerical value of its amplitude $I_{P,0}$.

Coil B is closed, so the voltage drop across it is zero:

$$\mathcal{E}_B(t) = -N_B \frac{d(k\Phi_A + \Phi'_B)}{dt} = -\mu\mu_0 g N_B \left(k N_A \frac{dI_P(t)}{dt} + N_B \frac{dI_B(t)}{dt} \right) = 0.$$

Hence we get

$$\frac{dI_P(t)}{dt} = -\frac{N_B}{k N_A} \frac{dI_B(t)}{dt}.$$

Let us use resultom, пораженным в parte C3:

$$-\frac{N_B}{k N_A} \frac{dI_B(t)}{dt} = \frac{dI_P(t)}{dt} = \frac{1}{\mu\mu_0 g N_A^2} \left(U_g(t) - \mu\mu_0 g k N_A N_B \frac{dI_B(t)}{dt} \right).$$

И we express value $\frac{dI_B(t)}{dt}$:

$$\frac{dI_B(t)}{dt} = -\frac{k U_g(t)}{\mu\mu_0 g N_A N_B (1 - k^2)}.$$

Substituting в expression for $\frac{dI_P(t)}{dt}$:

$$\frac{dI_P(t)}{dt} = \frac{U_g(t)}{\mu\mu_0 g N_A^2 (1 - k^2)} = \frac{U_g(t)}{L_A (1 - k^2)}.$$

Since мы знаем, что $U_g(t) = U_{g,0} \cos 2\pi f_0 t$:

$$I_P(t) = -\frac{U_{g,0}}{2\pi f_0 L_A (1 - k^2)} \cos 2\pi f_0 t.$$

Finally we get:

From here we get

$$\frac{dI_P(t)}{dt} = -\frac{N_B}{k N_A} \frac{dI_B(t)}{dt}.$$

Let's use the result obtained in point C3:

$$-\frac{N_B}{k N_A} \frac{dI_B(t)}{dt} = \frac{dI_P(t)}{dt} = \frac{1}{\mu\mu_0 g N_A^2} \left(U_g(t) - \mu\mu_0 g k N_A N_B \frac{dI_B(t)}{dt} \right).$$

And we express the value $\frac{dI_B(t)}{dt}$:

$$\frac{dI_B(t)}{dt} = -\frac{k U_g(t)}{\mu\mu_0 g N_A N_B (1 - k^2)}.$$

Substituting into the expression for $\frac{dI_P(t)}{dt}$:

$$\frac{dI_P(t)}{dt} = \frac{U_g(t)}{\mu\mu_0 g N_A^2 (1 - k^2)} = \frac{U_g(t)}{L_A (1 - k^2)}.$$

Since we know that $U_g(t) = U_{g,0} \cos 2\pi f_0 t$:

$$I_P(t) = -\frac{U_{g,0}}{2\pi f_0 L_A (1 - k^2)} \cos 2\pi f_0 t.$$

Finally we get:

Answer: \textbf{Answer:}

$$I_{P,0} = \frac{U_{g,0}}{2\pi f_0 L_A (1 - k^2)} = (4.6 \pm 1.0) \text{ mA}$$

C8**0.60 pts***Determine $I_{P,0}$ experimentally.*

Let's close coil B and determine the amplitude of the current in the primary coil $I_{P,0}$ using the method previously described in paragraph C5 (using a series-connected resistor). We get:

We get:

The obtained values coincide, taking into account the error.

Answer: \textbf{Answer: } $I_{P,0} = (5.0 \pm 0.3) \text{ mA}$.

C9**0.20 pts**

Express the self-inductance of series-connected coils L_{A+B} in the case when the fluxes from different coils are summed. Express your answer using L_A , L_B , k .

Let us express the total induced emf on series-connected coils by finding the total flux through coils A and B:

$$\mathcal{E}_{A+B}(t) = -N_A \frac{d(\mu\mu_0 g N_A I_A(t) + k\mu\mu_0 g N_B I_B(t))}{dt} - N_B \frac{d(k\mu\mu_0 g N_A I_A(t) + \mu\mu_0 g N_B I_B(t))}{dt}$$

$$I_A(t) = I_B(t) = I_{A+B}(t) \Rightarrow \mathcal{E}_{A+B}(t) = -(L_A + L_B + 2k\sqrt{L_A L_B}) \frac{dI_{A+B}(t)}{dt}$$

И по олимпиаде we get inductance L_{A+B} :

$$\mathcal{E}_{A+B}(t) = -N_A \frac{d(\mu\mu_0 g N_A I_A(t) + k\mu\mu_0 g N_B I_B(t))}{dt} - N_B \frac{d(k\mu\mu_0 g N_A I_A(t) + \mu\mu_0 g N_B I_B(t))}{dt}$$

$$I_A(t) = I_B(t) = I_{A+B}(t) \Rightarrow \mathcal{E}_{A+B}(t) = -(L_A + L_B + 2k\sqrt{L_A L_B}) \frac{dI_{A+B}(t)}{dt}$$

And by definition we get inductance L_{A+B} :

Answer: \textbf{Answer: }

$$L_{A+B} = L_A + L_B + 2k\sqrt{L_A L_B}.$$

C10**0.30 pts***Determine L_{A+B} experimentally.*

The measurement method is similar to the method in C5, only now the coils will be connected in series in a circuit.

$$U_{A+B,} = (6.1 \pm 0.1) \text{ V}, \quad I_{A+B,} = (0.50 \pm 0.05) \text{ mA}.$$

$$U_{A+B,} = (6.1 \pm 0.1) \text{ V}, \quad I_{A+B,} = (0.50 \pm 0.05) \text{ mA}.$$

Answer: \textbf{Answer: }

$$L_{A+B} = \frac{U_{A+B,}}{2\pi f_0 I_{A+B,}} = (190 \pm 20) \text{ mH}.$$

C11**0.20 pts**

Using the experimental values L_A , L_B , L_{A+B} , find the coupling coefficient k . Compare the resulting value with that obtained in point C5.

From the expression in C9 we express the coupling coefficient k :

Within the limits of error, the value coincides with that obtained in point C5.